REMANENT MAGNETIZATION OF SPILITE AND DIABASE IN THE FRANCISCAN FORMATION, WESTERN MARIN COUNTY, CALIFORNIA[1]

C. S. GROMMÉ AND H. J. GLUSKOTER[2]
Department of Geology and Geophysics, University of California, Berkeley

ABSTRACT

Recent detailed mapping of the Franciscan Formation in western Marin County, California, which includes three large bodies of spilitic pillow lavas and diabases, resulted in an interpretation of the structure as a northeast-dipping homocline, rather than a syncline as was previously thought. In order to test this interpretation by an independent method, oriented specimens were collected from twenty-five sites in the igneous rocks, and the directions and intensities of their remanent magnetizations were determined in the laboratory. After correction for structural dip of the homocline, the mean direction of magnetization had declination N. 74° E. and inclination 44° downward, with semiangle of the cone of 95 per cent confidence equal to 9.4°. Stability and thermoremanent origin of the observed magnetization is indicated by the convergence of magnetization directions after dip correction, by the significant divergence of the mean magnetization direction from that of the present field, and by the results of alternating-field and thermal demagnetization experiments. Removal by means of a.c. fields of secondary components of magnetization from diabase specimens from seven of the sites gave a mean direction having declination N. 78° E. and inclination 47° downward, not significantly different from that obtained using all data. The corresponding virtual magnetic pole is located in the Atlantic Ocean, at longitude 50° W. and latitude 26° N. This pole position is completely at variance with all upper Jurassic and Cretaceous paleomagnetic data from the northern hemisphere and may be due either to an obscure early deformation of the sampling area or to relatively rapid changes in the geomagnetic field. The uniformity of the magnetization directions confirms the homoclinal-structure interpretation and supports the conventional use of the configuration of pillows in sequences of pillow lavas as stratigraphic indicators.

INTRODUCTION

Purpose.—The object of this investigation was to employ paleomagnetic evidence to evaluate the reliability of the shapes of pillows as stratigraphic indicators in a sequence of spilitic pillow lavas and diabases in the late Mesozoic Franciscan Formation of the California Coast Ranges. Attempts to reconstruct the geologic history of the Franciscan Formation have been hampered by difficulties in field mapping arising from structural complexity, uniform lithology, lack of fossils, and limited exposures. Recent remapping in western Marin County, aided by extensive artificial exposures, has resulted in a structural interpretation fundamentally different from those previously published, and it appeared desirable to obtain independent evidence bearing on this. In addition, the new exposures provided an unusual opportunity to obtain paleomagnetic data for the first time from the Franciscan Formation.

Location and areal distribution.—The basic igneous rocks investigated occur in western Marin County approximately 20 miles north of San Francisco. This portion of the California Coast Ranges is divided into two separate and distinct blocks by the San Andreas rift zone. The western block consists of Cretaceous granitic rocks overlain by Miocene and Pliocene shales and sandstones. The much larger eastern block contains the spilites and diabases, and in this part of Marin County consists of a eugeosynclinal lithologic assemblage previously mapped as Franciscan by Lawson (1914) and Weaver (1949).

Figure 1 is an outline geologic map of the eastern block. The Franciscan Formation here consists predominantly of lithic and arkosic graywackes with interbedded shale, chert, and basic volcanic rocks and is intruded by diabase and serpentine. The basic igneous rocks are pillow lavas and massive lava flows in large part and, with

[1] Manuscript received October 19, 1963.

[2] Present address: Illinois State Geological Survey, Urbana, Illinois.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

the exception of the serpentine, are restricted almost entirely to three somewhat elongate bodies which trend between N. 30° W. and N. 40° W. and extend over a distance of 15 miles. As mapped, these bodies have maximum widths slightly greater than $1\frac{1}{2}$ miles. They appear to constitute a single and distinct stratigraphic unit, which is terminated on the north by an intensely disturbed zone of sheared graywacke containing numerous isolated blocks of various lithologies.

Structure.—The dominant structural trend in this area is northwest-southeast, paralleling that of the Coast Ranges in general. It is delineated not only by the outcrop pattern of igneous rocks but also by three other zones which have been mapped within the sedimentary rocks on the basis of presence or absence of detrital K-feldspar and the lithic or arkosic composition of the graywackes (Gluskoter, 1964). The sandstones east of the volcanic rocks differ markedly from those to the west. Between the San Andreas fault and the serpentine shown on the eastern part of the map (fig. 1), all the sedimentary and volcanic rocks appear to form a homocline dipping to the

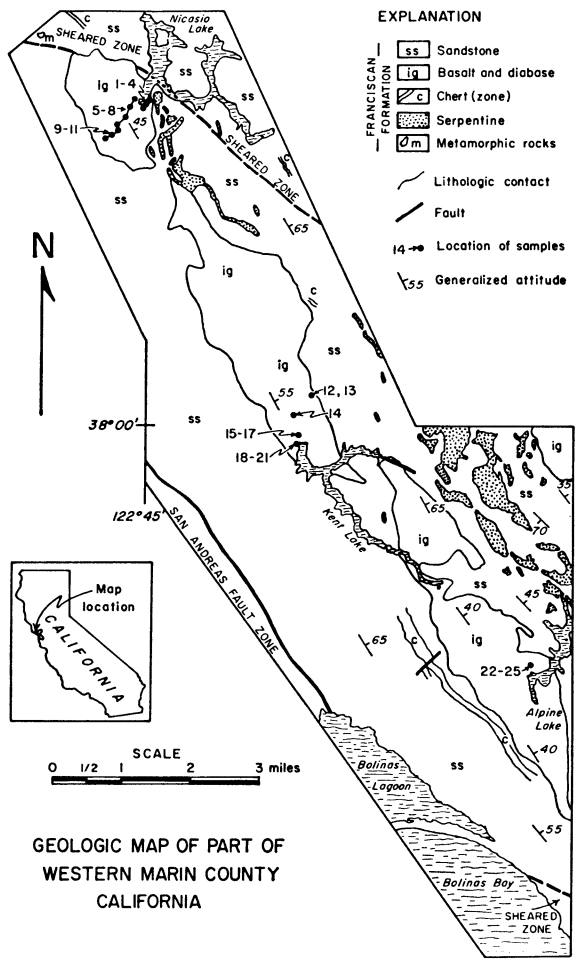


FIG. 1.—Outline geologic map of part of western Marin County, California, showing sampling localities in Franciscan spilitic basalt and diabase.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

northeast. The area is poorly exposed, and where bedding is visible in the massive graywackes, slumping often renders attitude measurements undependable. Recent dam and road construction has fortunately provided extensive artificial exposures in the lavas and diabases. The homoclinal-structure interpretation is based largely on these. The shapes of the pillows in the lavas and, where observable, graded bedding in the sandstones uniformly indicate tops of beds facing toward the northeast; thus the igneous and sedimentary zones mapped appear to be actual stratigraphic units. East of the southern volcanic body, the sediments are extensively intruded by serpentine and appear to form a syncline and anticline. The small volcanic area in the eastern corner of the map area is poorly exposed and is stratigraphically higher than the main bodies; it was not included in the present study.

The gross outcrop pattern of these igneous bodies is more complex than might be expected for a sequence of submarine flows simply folded about an apparently horizontal axis. The areas of graywacke about 2,000 feet in extent, which separate the three main volcanic bodies, may represent original positive areas on an irregular surface over which the lavas were extruded; they may be areas in which clastic sedimentation continued while three steep-sided volcanic piles were formed around separate eruptive centers; or they may be the result of faulting which has separated the original volcanic unit into three bodies *en échelon*. A fault might be indicated by the unusually lithic graywacke and a small amount of serpentine between the central and the southern bodies because these rocks are otherwise restricted to the northeastern part of the map area. However, there is no more direct evidence of faulting in that locality, and none at all between the central and northern bodies. Small embayments containing graywackes like those of the adjoining areas occur in the volcanic bodies; they may be the expressions of small folds with steep axes, but there is no other evidence of such folding. The very large embayment which nearly divides the southern body in two is almost certainly not the result of folding, and it is too abrupt to be explained as a stratigraphic feature; thus it must have been caused by faulting. This, however, cannot otherwise be demonstrated because of limited exposures. Numerous fractures and minor shear zones with small displacement are evident in the artificial exposures mentioned above.

Because previous mapping by Weaver (1949) had resulted in a structural interpretation at variance with the above, namely, that the northern and central volcanic bodies lay in the center of a syncline,[3] and because graded bedding in the graywackes is scarce, it was considered necessary to reexamine the reliability of pillow lavas as indicators of stratigraphic sequence. The pillows that comprise many of these flows are extremely variable in size and shape, and cross sections of them include a wide variety of forms. Their longest dimensions range from 6 inches to more than 6 feet and are from two to six times their shortest dimensions. Generally the two longer axes lie in the bedding plane (or define it), but there are equant individuals, and a few have their shortest axes in the bedding plane. Adjacent pillows are not connected; each pillow is separated from its neighbors by a highly sheared-appearing chloritic matrix. Despite the appearance of this matrix, there has been very little relative displacement of pillows except in the well-defined planar shear zones mentioned above, which cut through pillows and matrix alike. Plate 1 illustrates the shapes and sizes of some

[3] Lawson's (1914) section across the southern igneous body shows it as an intrusion with steeply dipping contacts, composed of diabase and basalt.

PLATE 1

SPILITE PILLOWS IN NICASIO CREEK SECTION

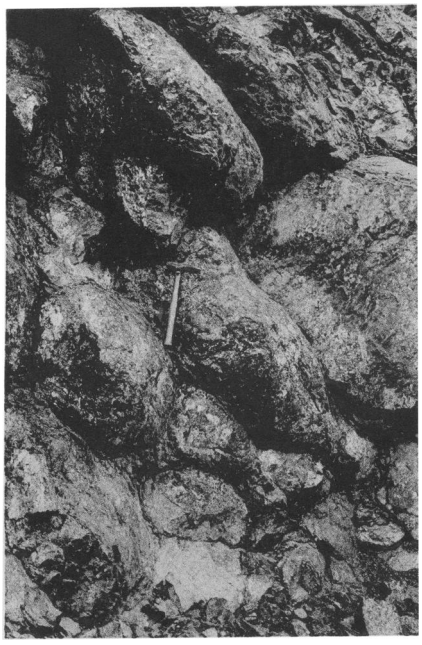
A, Tops facing toward viewer's left.

B, *C*, Tops facing toward viewer's right.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).



A

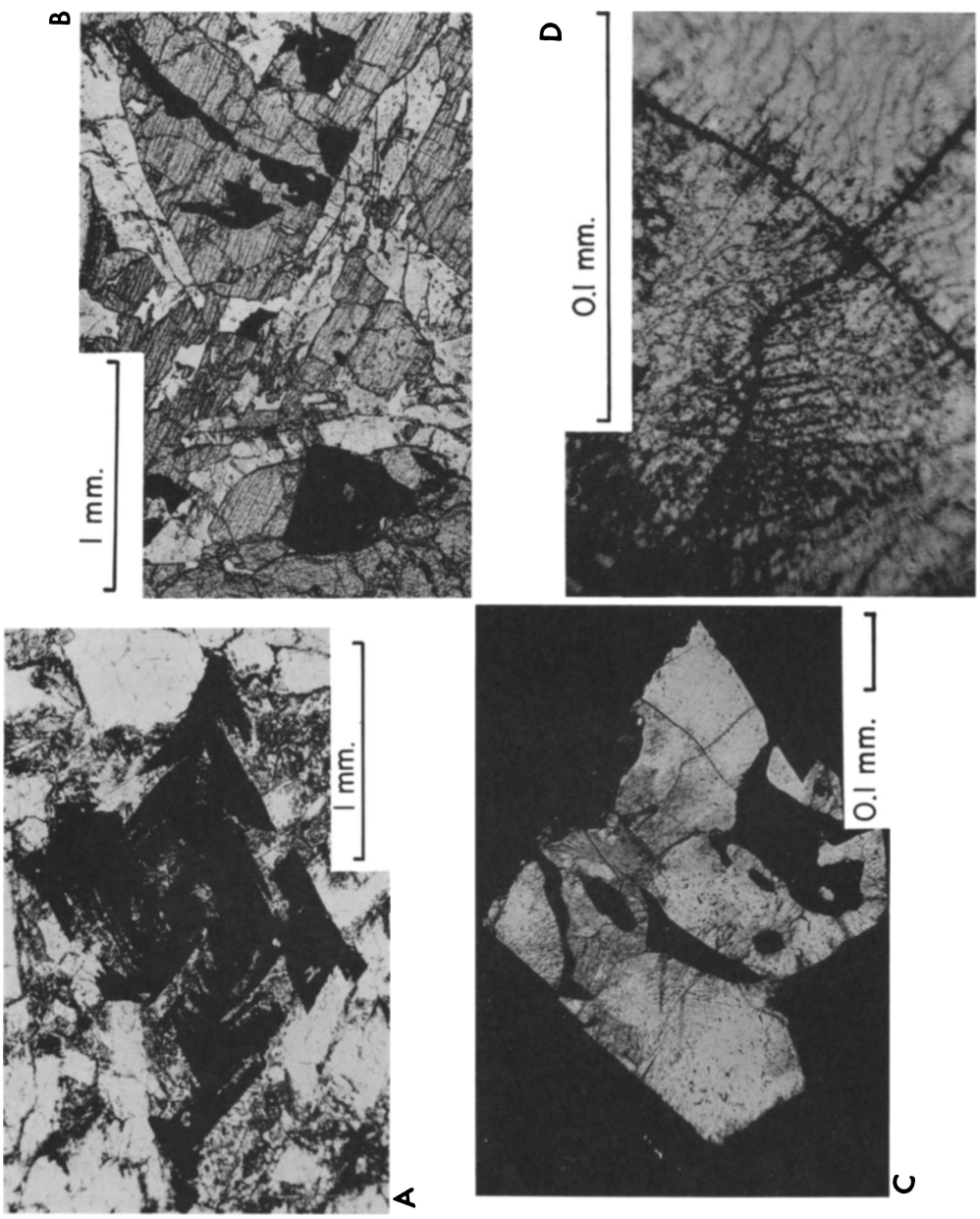


B



C

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).



This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

typical pillows, all of which are exposed in a new road cut which transects the northern body.

Shrock (1948, p. 364) has classified pillow shapes into the following five groups, all of which are represented in the area studied: (1) large tabular bodies having the shape of a mattress, with roughly parallel top and bottom; (2) bun-shaped pillows, or hemispheres with the flat side down; (3) flattened balloon-shaped pillows, each with a prominent cusp extending downward; (4) plano-convex forms similar to the mattress pillows except for a convex upper surface; (5) loaf-shaped pillows, having convex upper and concave lower surfaces. Shrock (1948, p. 359–367) has reviewed in some detail the arguments concerning the stratigraphic interpretation of pillow shapes. He concluded that they are probably reliable if used with care, that in general pillows are convex upward, and that the bottom surface of a pillow conforms to the upper surfaces of those beneath it, resulting in the downward-projecting cusps typical of types 3 and 5. Carlisle (1963) has recently described pillow lavas and pillow breccias on Quadra Island, British Columbia. Here the relations between close-packed pillow lavas, underlying limestones and limy cherts, and overlying breccias confirm Shrock's conclusions as given above.

In the area mapped, large flat pillows (types 1 and 4) were considered to lie in the bedding plane and thus to define it, and types 2, 3, and 5 were used to determine sequence. Most observations support the homoclinal interpretation; overturned beds were found only in the vicinity of sites 20–21 (fig. 1), and these also face northeast. As suggested by Shrock (1948, p. 366), variole or amygdule concentrations did not prove as consistently useful in determining sequence, since they may be distributed uniformly around the periphery of a pillow. In summary, the bedding in the igneous bodies, as defined by large, flat pillows and contacts between flows, is parallel to that in the graywackes; all dips are eastward except in the small area of overturned beds and in the folds east of the igneous bodies; and the tops of all beds appear to face toward the northeast except in the latter area, in part of which graded bedding dips 70° southwest (fig. 1) and probably faces southwest. Thus it appears that no fold exists in the igneous bodies, or if one does, it must be isoclinal.

Petrography of the basic igneous rocks.— The pillow lavas and massive flows are for the most part extremely fine-grained and are spilitic; that is, the plagioclase in them is albite. Diabasic rocks, coarser but otherwise mineralogically similar, are associated with them either as interiors of thick flows or as separate intrusive bodies. Rocks sufficiently coarse-grained to be studied petrographically consist primarily of the following minerals:

1. Albite: An_0–An_8, $2V_Z = 80°$–$88°$, determined by universal stage; $\beta = 1.534 \pm .002$; $2\theta(131) - 2\theta(1\bar{3}1) = 1.1°$–$1.3°$ and $2\theta(\bar{1}32) - 2\theta(131) = 2.65°$–$2.75°$, indicating the low-temperature structural state (Smith, 1956).
2. Augite: iron-rich; $2V_Z = 50° \pm 2°$; $\beta = 1.702 \pm .003$; generally unaltered, somewhat pink in thicker sections, ophitically intergrown with albite laths in the diabases.
3. Chlorite: used in the broadest sense as referring to green minerals with moderately high refractive indices ranging from 1.649 to 1.651 and with low birefringence. The pre-

PLATE 2

MAGNETITE IN ALBITE DIABASE

A, Skeleton crystal from site 19 with albite, chlorite, and a little relict pyroxene; plane-polarized transmitted light.

B, Intergrowth in pyroxene from site 18 with albite and minor chlorite; plane-polarized transmitted light.

C, Typically altered crystal from site 2; magnetite is light gray, sphene is dark gray, plagioclase and pyroxene are black; reflected light.

D, Portion of the same field as *C*, showing details of the sphene veinlets.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

dominant chlorite gives an extremely well-developed 7-Å peak and a less well developed 14-Å peak on X-ray diffraction patterns.

4. Magnetite: appears in thin sections as opaque skeleton crystals or as irregular intergrowths, usually in pyroxene (pl. 2, *A*, *B*); often "milky" in oblique reflected light. Polished surfaces in vertically incident light show that these opaque-appearing crystals are mostly cores of homogeneous magnetite surrounded by material, apparently an alteration product, with reflectivity intermediate between the magnetite and the albite and pyroxene and containing fragments of magnetite. Alternatively, the magnetite grains may be permeated throughout by veinlets of this material, ramifying from the margins and from fractures (pl. 2, *C*, *D*). X-ray diffraction patterns of two ferromagnetic concentrates obtained with a hydraulic magnetic elutriator from diabase ground to pass 270 mesh indicate magnetites containing about 5 per cent of the ulvöspinel end-member, assuming stoichiometry (Nagata, 1961, p. 82). Much sphene was also identified in the concentrates. This is the alteration product seen in the polished sections and is too fine to be removed during the concentration process. A trace of ilmenite is present in one concentrate, although none was observed in the polished sections.

Two igneous rocks which do not contain albite occur with the spilites and albite diabases: they are a labradorite diabase and a bytownite-bearing basalt, neither of which could be traced more than a few hundred feet.

MAGNETIC PROPERTIES OF THE BASIC IGNEOUS ROCKS

Objectives and methods.—A paleomagnetic investigation of these rocks was undertaken (1) to provide independent evidence of the presence or absence of a possible isoclinal (or other) fold in the volcanic bodies and thus (2) to evaluate further the consistency of pillow shapes, and (3) to detect any general lack of uniformity of structure that had not been found through detailed mapping, such as relative rotations of the three volcanic bodies. Provided it could be shown that the natural remanent magnetization was of primary origin, uniformity of magnetization directions within and between the three bodies would indicate that there had been little or no deformation other than that observable through field mapping. However, if the magnetization directions were divergent in some systematic way, this might be interpreted in terms of additional folding and/or faulting.

A total of 127 oriented specimens was collected at twenty-five sites, all of which were in artificial exposures. The site locations are shown in figure 1. An attempt was made to obtain a uniform stratigraphic sample, within the limitations of the exposures available. As is apparent from figure 1, this was most nearly accomplished in the northern body.

The specimens are cylinders 0.98 inch in diameter and 1 inch long. Specimens from sites 1–10, 18, 19, and 21 were cored in the field using a portable, hand-held, gasoline-powered diamond drill; the remainder were cored on a drill press in the laboratory from accurately oriented hand specimens. Specimen spacing at individual sites ranged from zero to several feet as dictated by local conditions.

Directions and intensities of magnetization were measured with a motor-driven induction magnetometer of the type described by Nagata (1961, p. 61–62, figs. 2–17 and 2–18). This particular instrument has a lower limit of useful sensitivity about 2×10^{-6} e.m.u.; measurements below about 10^{-5} e.m.u. have uncertainties of ± 5 degrees in direction and ± 20 per cent in intensity. Measurements above 10^{-4} e.m.u. are reproducible to $\pm \frac{1}{2}$ degree in direction and ± 2 per cent in intensity. Reversible weak-field susceptibilities were measured with a commercial bridge in a 1,000 c.p.s. alternating field of about 1 oersted peak intensity.

Results.—The directions of magnetization of all specimens are shown in figures 2 and 3. In these figures, the directions shown in the second of each pair of projections are obtained from those in the first by rotation through the angle of dip of the bedding at each site about a horizontal axis correspond-

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

ing to the strike of the bedding. This is not the only permissible method of structural restoration, but it is the simplest and most obvious, and it is justified by the general lack of steep fold axes here and elsewhere in the Coast Ranges. The magnetization directions show a rather large scatter which is not diminished much by the structural restoration. It is apparent, however, that there is no significant difference between the groups of magnetization directions from the three volcanic bodies after structural restoration. Furthermore, both before and after restoration the directions diverge significantly from the present direction of the geomagnetic field and from the field direction which would correspond to a dipole directed along the axis of the earth's rotation.

Figure 3, *c*, *d* shows the mean directions of magnetization at each site, calculated as

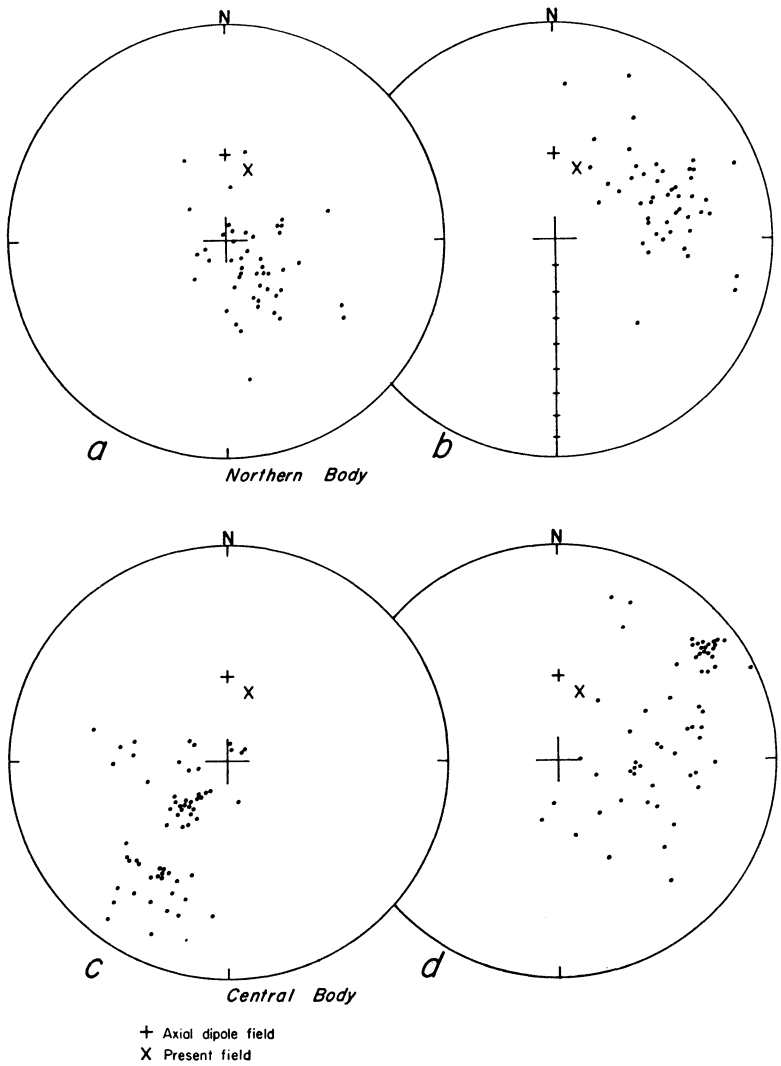


FIG. 2.—Equal-area projections of magnetization directions, north-seeking poles plotted, all on lower hemisphere; (*a*) and (*c*), directions observed in all specimens from northern and central bodies, respectively; (*b*) and (*d*), same data after restoring bedding to horizontality.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

the vector sum of specimen directions assigning unit magnitude to each (Cox and Doell, 1960, p. 669). Also shown are the over-all mean direction and the mean of the site-mean directions, calculated in the same manner. These data are included in tables 1, 2, and 3, respectively, together with the relevant statistical parameters. It can be seen from table 1 that the number of specimens at each site is small, with the exception of site 20 from which many specimens of an exceptionally fresh and strongly magnetized diabase were collected for demagnetization experiments. Site 20 is thus the main cause of the divergence between the over-all mean and the mean of the site means. It will be

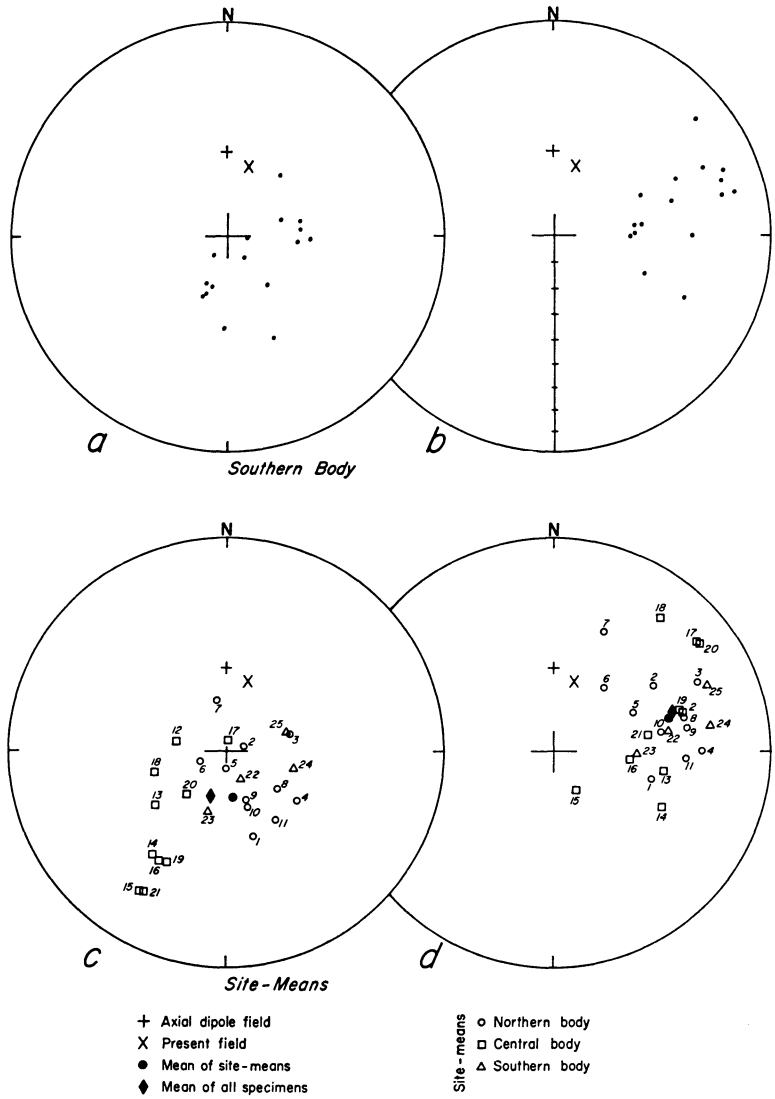


FIG. 3.—Equal-area projections of magnetization directions; north-seeking poles plotted, all on lower hemisphere; (*a*), directions observed in all specimens from southern body; (*c*), calculated means of observed directions at each site, together with over-all mean directions calculated as the mean of the site means and as mean of all specimens, (*b*) and (*d*), same data after restoring bedding to horizontality.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

TABLE 1

SITE-MEAN DIRECTIONS OF MAGNETIZATION

SITE	N*	OBSERVED*			RESTORED*			α_{95}*	BEDDING	
		D	I	k	D	I	k		Dip	Strike
1.....	3	163	55	20	106	51	.....	28	40° NE.	N. 40° W.
2.....	7	73	83	61	56	44	.....	8	40° NE.	N. 40° W.
3.....	5	76	65	84	64	27	.....	8	40° NE.	N. 35° W.
4.....	5	126	57	16	90	32	.....	20	40° NE.	N. 35° W.
5.....	6	180	83	56	64	56	57	9	37°/40° NE.	N. 35°/45° W.
6.....	3	248	80	53	38	59	.....	17	40° NE.	N. 40° W.
7.....	4	349	70	16	22	39	14	25	40°/38° NE.	N. 50°/55° W.
8.....	3	128	66	180	75	38	.....	10	45° NE.	N. 45° W.
9.....	4	159	70	82	80	37	.....	10	55° NE.	N. 35° W.
10.....	5	160	67	35	80	48	44	12	45°/50° NE.	N. 45° W.
11.....	3	145	58	270	93	39	.....	8	45° NE.	N. 40° W.
12.....	5	281	71	59	72	38	.....	10	70° NE.	N. 10° W.
13.....	7	232	57	10	100	47	.....	19	70° NE.	N. 10° W.
14.....	6	214	40	24	118	43	.....	14	80° NE.	N. 15° W.
15.....	7	211	25	33	151	73	.....	11	60° NE.	N. 40° W.
16.....	5	210	40	2000	96	61	.....	2	70° NE.	N. 35° W.
17.....	5	10	86	116	52	17	.....	7	70° NE.	N. 35° W.
18.....	5	253	62	13	38	22	.....	22	85° SW.†	N. 40° W.
19.....	3	208	41	70	72	38	.....	13	85° SW.†	N. 40° W.
20.....	17	222	68	250	53	16	.....	2	85° SW.†	N. 40° W.
21.....	3	210	25	57	80	54	.....	15	85° SW.†	N. 40° W.
22.....	3	152	78	44	80	45	.....	22	45° NE.	N. 30° W.
23.....	5	196	66	110	92	59	.....	7	45° NE.	N. 30° W.
24.....	4	104	64	14	81	28	.....	27	45° NE.	N. 30° W.
25.....	4	74	68	29	66	24	.....	16	45° NE.	N. 30° W.

* N, number of specimens at each site; D, declination, degrees east of north; I, inclination, degrees downward; k, estimated precision parameter [$k = (N-1)/(N-R)$, where R is the vector resultant of all measurements at each site, assigning unit magnitude to each specimen]; α_{95}, semiangle of cone of 95 per cent confidence.

$$\cos \alpha = 1 - \frac{N-R}{R}\left[\left(\frac{1}{P}\right)^{1/(N-1)} - 1\right],$$

where $P = 0.05$.

† Overturned beds.

TABLE 2*

OVER-ALL MEAN DIRECTION OF MAGNETIZATION, WITH STATISTICAL PARAMETERS CALCULATED ACCORDING TO THE TWO-LEVEL SCHEME OF WATSON AND IRVING (1957)

	D	I	N'	B	$\overline{W}$	ω	β	k'	α_{95}
Observed....	197	72	127	25	5.02	35	10	237	9.1
Restored.....	71	42	127	25	5.02	35	12	281	8.4

* D, declination, degrees east of north; I, inclination, degrees downward; N', total number of specimens; B, number of sites; $\overline{W}$, weighted average of number of specimens obtained from each site; ω, estimated within-site precision parameter; β, estimated between-site precision parameter; k', estimated over-all precision parameter; α_{95}, semiangle of cone of 95 per cent confidence.

TABLE 3*

MEAN OF SITE-MEAN DIRECTIONS OF MAGNETIZATION

	D	I	N	R	k	α_{95}
Observed.....	172	72	25	22.28	8.8	10.3
Restored.....	74	44	25	23.13	12.8	8.4

* Symbols as in table 1, except that N is the number of sites and R is the vector resultant of all site means, assigning each site mean unit scalar magnitude.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

shown subsequently that the directions at this site contain an appreciable secondary component, and for this reason the mean of site means (table 3) is considered to represent more accurately the true average direction. Because the physical distribution of the samples is appropriate to the two-level statistical analysis of Watson and Irving (1957), the results of such an analysis are given in table 2 as they apply to the over-all mean direction. Watson and Irving's analysis takes fullest advantage of the data available; however it is strictly applicable only in cases where the within-site precision is uniform for all sites; and, for the present data, k for individual sites ranges from 10 to 2,000 (table 1). A comparison of the values of β in table 2 with those of k in table 3 shows that the estimate of the between-site precision of the mean is not greatly affected by the choice of one or the other method of calculation.

The values of N and R (table 1) have been used to calculate corresponding values of α_{95}, the semiangle of the circular cone within which the true site mean may be expected to lie with 95 per cent probability (Fisher, 1953). It should be noted that the validity of the use of Fisher's precision parameter k for purposes of comparison does not necessarily depend on the conformity of the statistical distribution of directions to Fisher's model, but the validity of the two-level scheme of analysis and of the 95 per cent confidence interval α_{95} are so dependent. In this particular case the conformity of data to Fisher's distribution is indeterminate for individual sites since N is generally small (table 1); the total data after structural restoration and the site means do not appear to depart markedly from it (figs. 2 and 3). Although the values of α_{95} given in table 1 have been calculated by the formula which is exact for N as small as 2, they are included only to show the degree of uncertainty in determination of bedding attitude that is tolerable in terms of the magnetic results. Increasing N at sites with low k would of course reduce this tolerance; with the present sampling scheme it appears that the uncertainties in D and I and in dip and strike are of roughly the same magnitude at many of the sites. The bedding attitudes used in obtaining the restored directions at each site have also been included in table 1. These are the averages of many measurements by two observers; individual measurements have accuracies ranging from $\pm 2°$ to $\pm 5°$, depending on the flatness of large pillows or of groups of pillows.

Structural significance.—Although the structure of these rocks is essentially homoclinal, as stated above, sufficient variation in attitude exists to suggest the applicability of Graham's (1949) fold test. This well-known test for the stability and pre-deformational origin of magnetization of rocks gives a positive result when scattered magnetization directions in folded rocks converge upon geometrical unfolding. Such a convergence is not readily apparent in figures 2 and 3; however β and k' in table 2 and k in table 3 are all significantly greater for the restored directions. This degree of convergence of site means after unfolding is taken to be a statistically positive result of Graham's fold test. Furthermore, although the bedding attitudes at each site are uniform for most sites, at sites 5, 7, and 10 large specimen separations exist and, accordingly, two attitudes are given in table 1. For site 7, k decreases slightly after unfolding, and for site 5 the increase in k is insignificant, but for site 10 k increases from 35 to 44. Additional evidence for the positive result of the fold test was obtained from demagnetization experiments to be described below.

The between-site scatter still remains considerably greater than the average within-site scatter after unfolding (table 2), which may be due at least in part to tectonic causes, in part to geomagnetic secular variation, and in part to variable degrees of secondary magnetization. The latter two possibilities will be discussed below in connection with the demagnetization experiments. The between-site scatter could of course be reduced to nil by other structural restora-

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

tions, for example, rotation of individual site means about various non-horizontal axes or about axes not parallel to bedding, multiple rotations corresponding to two or more hypothetical deformations, or allowance for possible initial dips in the lavas. Such restorations would have to be based on the magnetic data alone, and the choice of rotations would thus be so free as to provide no unambiguous conclusions about the deformational history of the rocks.

It is evident at this point that the homoclinal-structure interpretation and the stratigraphic consistency of pillow shapes are supported by the magnetic data, provided one assumes that the natural remanence (NRM) is thermoremanent magnetization (TRM) acquired in the ambient field during the initial cooling of the igneous rocks. This assumption will be examined below. Since, however, it appears that the geomagnetic field has reversed its polarity an unknown but probably large number of times in the geologic past (Cox and Doell, 1960), the normal sense of polarity of these rocks has no bearing on whether or not this entire stratigraphic section is upright. The field evidence (graded bedding in the graywackes, pillow shapes in the lavas) is considered sufficient to demonstrate that it is upright in the light of the consistency of the magnetic data.

Two alternate explanations for the divergence of the observed and restored directions from the present geomagnetic field and from the axial dipole field can be offered: movement of the apparent magnetic pole, or tectonic rotation of the entire map area. For example, it can be seen from figure 3, *d* that a clockwise rotation of the rocks as a unit through some 70° about a vertical axis could account for all but about 10° of the divergence. Such a rotation is of the correct sense to have resulted from a hypothetical shear couple corresponding to the sense of movement (right-lateral) along the San Andreas fault, and it could have occurred either before or after the folding.[4] Post-folding rotation appears highly unlikely because of the parallelism of the strike of this homocline with the trends of fold axes in the Coast Ranges in general (see, for example, Jennings and Burnett, 1961). The folding antedates at least part of the displacement along the fault, but structures such as deflected fold trends which might be associated with the displacement are not common outside the fault zone itself. It could be argued that the rotation was pre-folding and was associated with the formation of the sheared zones which terminate the block on the north and south (see fig. 1); but although their temporal relationship to the San Andreas fault is obscure and little else is known about them, these zones appear also to truncate folds and thus to be post-folding, at least in part.

For the sake of argument, one can show (e.g., stereographically) that, starting with the bedding plane horizontal and the direction of magnetization within 10° of the axial dipole field, the present configuration of bedding and magnetization (as in fig. 3, *c*) can be obtained by a single clockwise rotation of about 90° around an axis plunging 60° and trending N. 80° W. As with the previous hypothesis, this is possible because the angle between the bedding plane and the direction of magnetization is within 10° of the downward inclination of the axial dipole field at this latitude. However, there is no direct geologic evidence in support of this rotation either, and it is clear from these simplest of the multitude of possible alternatives that postulating a tectonic origin for the magnetic divergence creates more geologic problems than it solves. Thus in spite of the lack of supporting evidence from the Jurassic or Cretaceous of North America wandering of the magnetic pole or predominance of the non-dipolar components of the geomagnetic field may be more plausible hypotheses. This will be discussed in greater detail below.

[4] In this particular case the two rotation operations (folding and rotation about a vertical axis) are commutative, the special conditions being that one axis is fixed in space while the other is fixed with respect to the rotated body, and that the angular relation between the two axes remains fixed.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

Stability and origin of natural remanence.—Figure 4 shows the distribution of intensities of natural remanence (J_n) and reversible weak-field volume susceptibilities (χ). The intensities fall into two well-defined groups, with the diabases being roughly 100 times more strongly magnetized than the basalts. This is not the usual relation in non-spilitic diabase-basalt associations; there the remanent intensities of both rock types are commonly within the same order of magnitude. For example, Creer (1962*a*) shows that for many samples from the Serra Geral Formation of Uruguay and Brazil the average remanence and the average susceptibility are about twice as great in the diabases as in the basalts. The peculiar distribution of susceptibilities for the spilitic basalts seen in figure 4 is due to overriding of the ferromagnetic susceptibility of magnetite by the paramagnetic susceptibility of the iron-bearing silicate minerals present, which has been calculated from chemical analyses of these rocks to be of the order of 5×10^{-5} at 25° C. If one could subtract the paramagnetic contribution, the ferromagnetic susceptibility of the basalts would probably turn out to be distributed much as the natural remanence. This was not attempted because the rocks are not chemically homogeneous and because the accuracy of the bridge is very low in this range.

The difference by two orders of magnitude in both the remanences and the susceptibili-

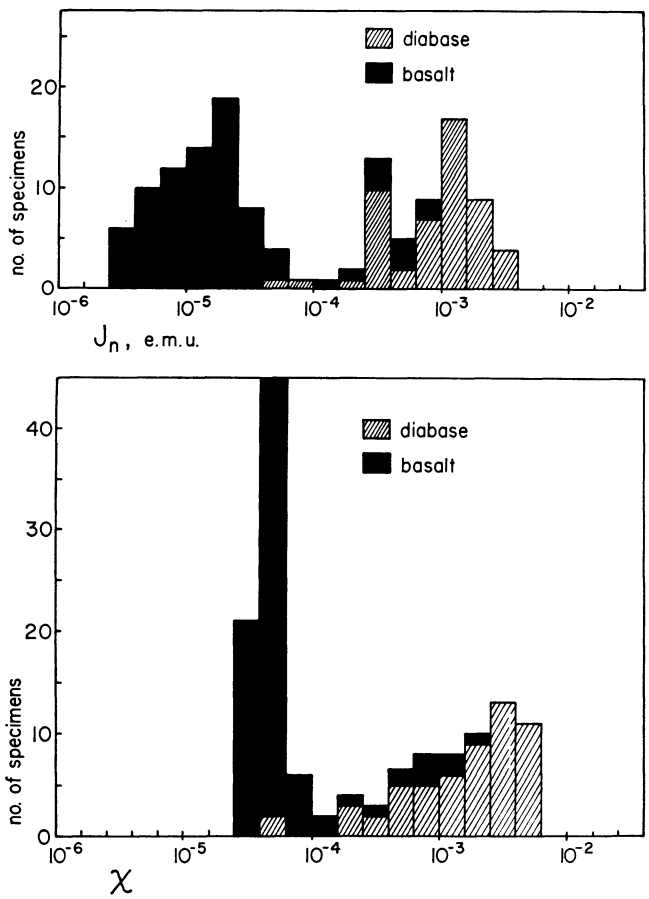


FIG. 4.—Histograms showing the distribution of values of remanent intensity of magnetization J_n and reversible weak-field susceptibility χ. Logarithm of the class interval is 0.2.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

ties of the diabases and the basalts can be correlated roughly with the actual difference in the content of magnetite in the two rock types as seen in thin sections. This difference may be primary, i.e., an initial result of magmatic crystallization. On the other hand, the occurrence of sphene in the diabases, associated with the magnetite and to all appearances an alteration product of it, suggests that the basalts may once have contained more (titaniferous) magnetite which was altered during the latter stages of cooling to a mixture of sphene and probably chlorite. A possible difficulty with this hypothesis is that in order to lower the saturation magnetization of the rock (and thus the other magnetic parameters), the trivalent iron in the magnetite must be reduced to the divalent state if the major iron-bearing product is to be chlorite. Alternatively, an epidote mineral could accommodate the trivalent iron; little has been observed, but its presence in the fine-grained rocks cannot be ruled out. Among the rocks chemically analyzed, no consistent difference in the Fe_2O_3/FeO ratio exists between the diabases and the basalts, although this ratio is quite variable.

Alteration of titanomagnetite to sphene with associated chlorite has recently been described in Precambrian basic rocks (nonspilitic and otherwise unmetamorphosed) by Desborough (1963). He has shown that, because relict ilmenite lamellae in the titanomagnetite have remained undistorted, the alteration is a constant-volume reaction and has concluded that a net loss of iron from the altered rocks must have occurred since the chlorite present could not account for the total iron in the original titanomagnetite. After reviewing the pertinent literature, Desborough (1963) further concluded that this type of heterogeneous alteration is due to fluids derived from the basic magma itself, and that it is not the result of subsequent regional metamorphism.

To obtain further evidence bearing on the origin and stability of the observed magnetization, alternating-field (a.c.) and thermal demagnetization experiments were carried out on some of the more strongly magnetized diabase specimens. A.C. demagnetization was done by tumbling the specimen about three axes simultaneously in a 60 c.p.s. alternating field which was continuously and smoothly reduced from its initial strength to zero over a period of two minutes by means of a motor-driven variable inductance. Apparatus of this type has been described by Creer (1959), Irving, Stott, and Ward (1961), and Cox (1961). Thermal demagnetization was done in a non-magnetic, non-inductive oven containing a nitrogen atmosphere, in a space field-free to less than 1 per cent of the earth's field.

The results of the demagnetization experiments are shown in figures 5, 6, 7, and 8. The changes in direction shown in figure 5 are systematic for specimens from sites 16, 17, and 20 but appear to be random for those from site 18. Data for a.c. demagnetization from site 18 are plotted in figure 5, *a* as an average of three specimen directions for each field increment; the individual directions wander about much more freely and in just as random a fashion, and the scatter between them increases with each field increment. In marked contrast are the results from site 20, which show steady and uniform changes in direction along great circles passing nearly through the present field direction. The direction approached is almost coincident with the over-all mean after structural restoration, but not before. This is interpreted to mean that the direction approached during demagnetization is the original stable direction of magnetization and that the rock acquired a secondary component of magnetization along the direction of the earth's field subsequently to folding. From figures 5, *b* and 7 it can be seen that the primary magnetization is stable in direction up to 500° C., and that the secondary magnetization is only a little less stable, since part of it persists up to at least 450° C. The origin of such a stable secondary magnetization is not known; lightning can be ruled out because electric storms are infrequent in the area and the exposure, which is well sheltered, is only a

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

few years old, and because the secondary component appears to be directed along the present field, which is not typical of lightning-struck rocks (Cox, 1961; Graham, 1961). The thermal stability of the primary component indicates that it is probably TRM.

earth's field is directed some 90° away from the mean end point after demagnetization for sites 16 and 20 (before restoration); thus the primary direction may lie beyond these end points and may not have been reached.

Demagnetization results for site 17 are somewhat variable. These specimens seem

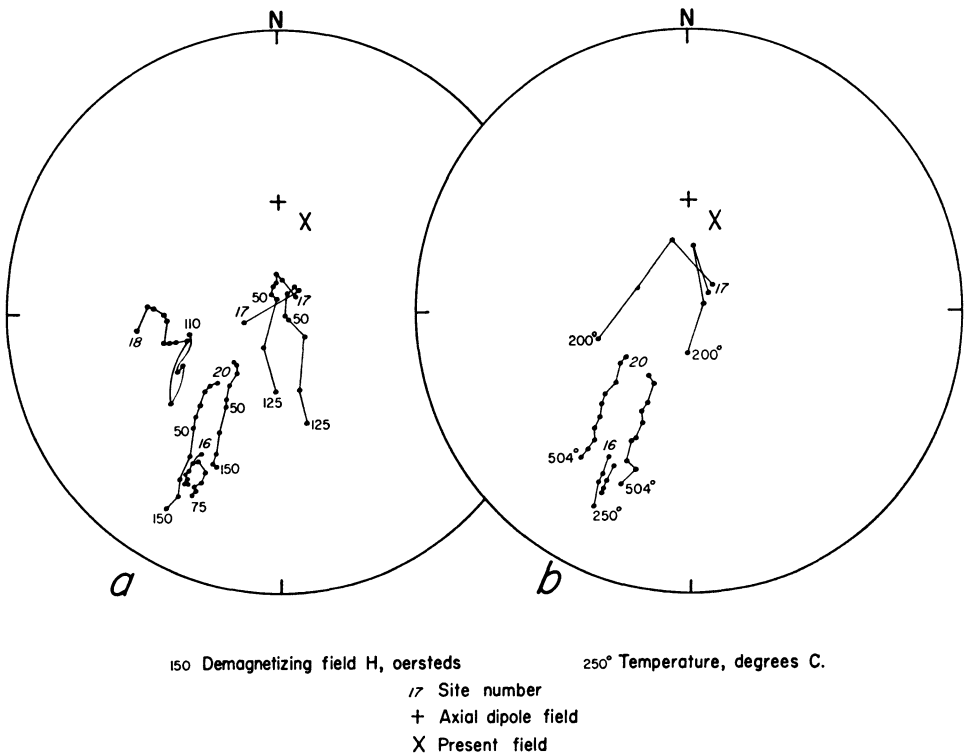


FIG. 5.—Equal-area projections of changes in direction of natural remanence during partial demagnetization; north-seeking poles plotted, all on lower hemisphere; site number is near initial direction; figure indicating maximum treatment is near final direction; (*a*), a.c. demagnetization; (*b*), thermal demagnetization; measurements made at room temperature; sites 16, 17, 20: individual specimens; site 18: average of three specimens.

The results for site 16 are similar to those for site 20, except that the secondary component is weaker and much less stable; the change is away from the earth's field before restoration, but is also away from the mean after restoration. The initially positive slopes of the demagnetization curves for site 16 and the inflections around 50 oe. and 200° in those for site 20 (figs. 6 and 7) indicate that the secondary component makes an obtuse angle with the primary component. The to carry three components of magnetization: the most stable one lies near the over-all mean; that with intermediate stability appears directed along the present field; the least stable and weakest one has unknown but probably random directions.

A clear-cut comparison of the relative effectiveness of the two methods of magnetic cleaning is difficult in the present case. With regard to directions (fig. 5), a.c. demagnetization seems to have given more consistent

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

results for site 17, while thermal cleaning was a little more effective for sites 16 and 20. In comparing figures 6 and 7, it appears that thermal demagnetization has separated the two components of magnetization a little better in the case of sites 16 and 20; for example, compare the respective slopes of the demagnetization curves for site 16 at 100 oe. and at 200°. The irregularities in the dashed parts of most of the a.c. demagnetization curves are associated with random changes in direction. This phenomenon has been observed by a number of workers and has been discussed in detail by Irving *et al.* (1961). The irregularities above 350° in the thermal demagnetization curve for site 17 (fig. 7) may be due to fluctuations in the residual field in the oven, although other specimens run simultaneously exhibited quite systematic behavior.

Because of the distinct possibility that the site-mean directions as given in table 1 and figure 3 may be influenced by various degrees of secondary magnetization, the a.c. demagnetization experiments were extended to include all specimens with sufficiently high intensities of natural remanence. The optimum demagnetizing field for specimens

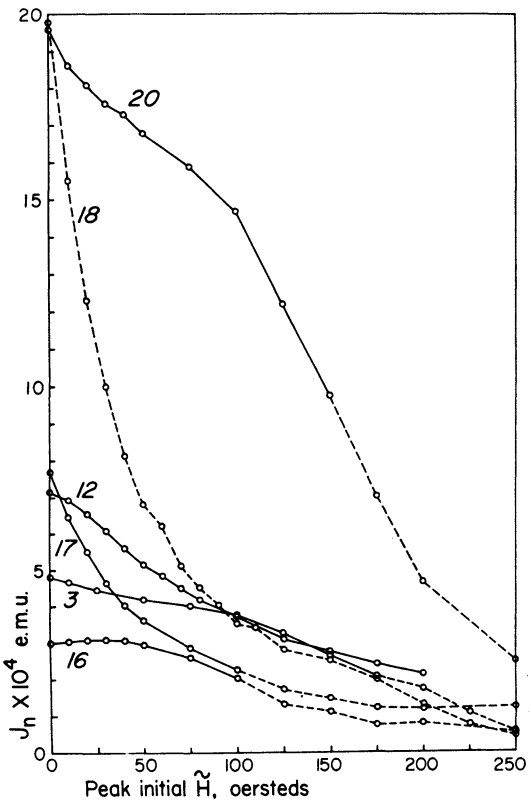


FIG. 6.—Change in intensity of natural remanence during a.c. demagnetization. Solid portions of curves correspond to data in figure 5, *a;* higher demagnetizing fields (*dashed portions of curves*) produce random changes in direction. Points plotted represent the averages of four specimens for site 3, five specimens for site 12, two specimens for site 16, two specimens for site 17, three specimens for site 18, and three specimens for site 20.

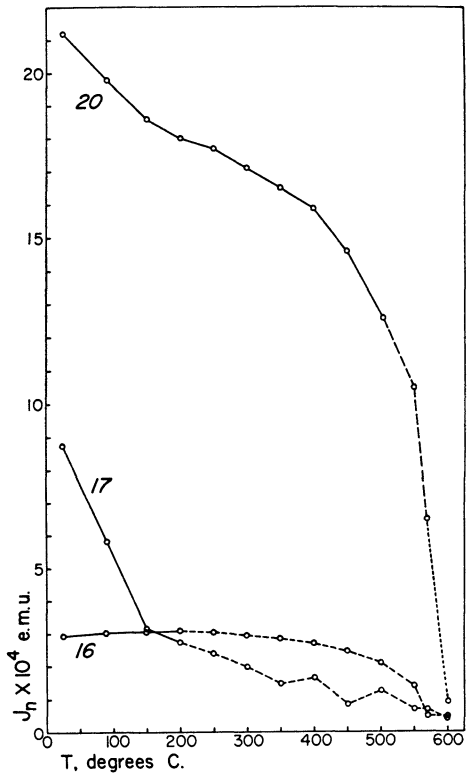


FIG. 7.—Change in intensity of natural remanence during thermal demagnetization (measurements made at room temperature). *Solid portions of curves* correspond to data in figure 5, *b;* higher temperatures (*dashed portions of curves*) produce random changes in direction; *dotted portion of curve* for site 20 indicates uncertainty in the Curie temperature region. The points plotted represent the averages of two specimens for site 16, two specimens for site 17, and six specimens for site 20.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

from each site was found by trial and error and was considered to be the maximum field which could be applied without producing random changes in direction. The results are given in table 4 and figure 8. Directions of magnetization at site 4 changed randomly during demagnetization, as did those at site 18 (fig. 5, *a*); these directions are not included. All the remaining sites are in spilitic basalt and have intensities of magnetization less than 10^{-4} e.m.u., and no demagnetization was done on specimens from them.

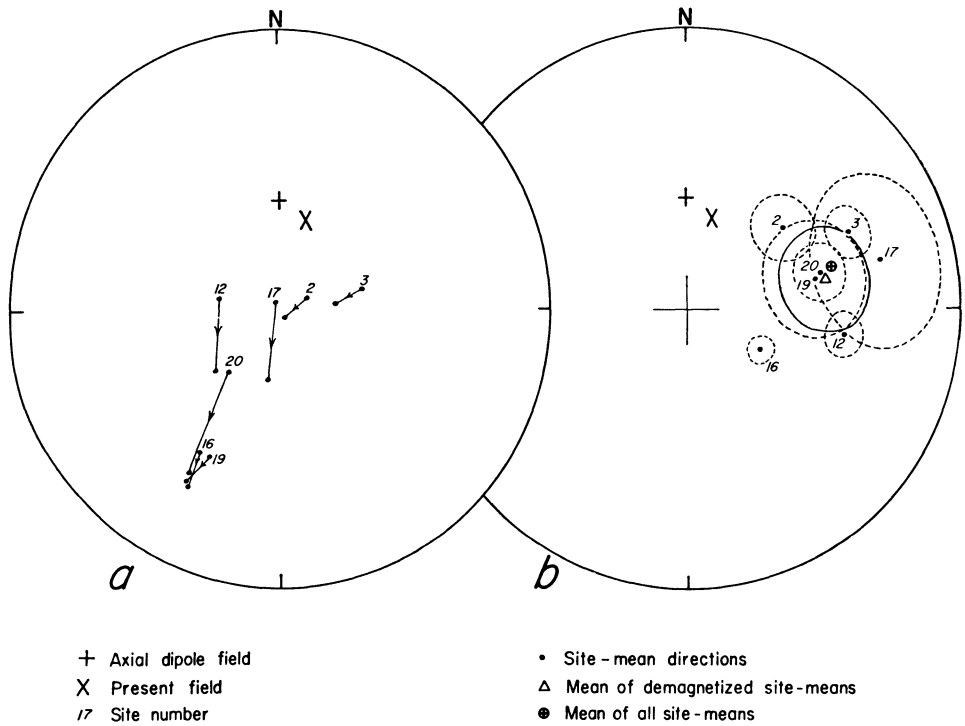


FIG. 8.—Results of all a.c. demagnetization; north-seeking directions plotted on lower hemisphere of equal-area projection. (*a*), mean directions of magnetization at seven sites, before demagnetization (adjacent to site numbers) and after demagnetization in various field strengths as given in table 4; (*b*), site-mean directions after demagnetization and rotation of bedding into horizontality, together with mean of demagnetized site means (table 4) and mean of all site means prior to demagnetization (table 3). *Circles*, 95 per cent confidence, are also included; *solid circle*, mean of demagnetized site means (table 4).

Table 4 shows that k increased significantly (i.e., the scatter was reduced) after demagnetization for sites 2, 3, and 12 but that the scatter was increased at sites 16, 17, 19, and 20. The increase in scatter is most pronounced for site 20; it appears to result from incompleteness of removal from some specimens of the secondary component, which was shown above to be unusually stable at this site. It may be worth noting that these latter four sites are all in a restricted area in the central body, while site 12 is in basalt some distance away and sites 2 and 3 are in the northern body (cf. fig. 1). It is clear, however, from the behavior of individual specimens from all sites during demagnetization that in every case a secondary component was systematically removed. It appears from figure 8, *a* that in the case of sites 2, 3, and 12 this component is not parallel to the present field.[5]

[5] The site-mean directions before demagnetization given in table 4 do not correspond in all cases to those in table 1 because some specimens from these sites were either too weakly magnetized or had been used for other purposes.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

The site-mean directions after demagnetization and structural restoration are shown in figure 8, *b*, together with the circles of 95 per cent confidence which are the same as those given in table 4, since the bedding attitudes are uniform at each of these sites. Also shown are the mean of the site means and the corresponding 95 per cent confidence circle, calculated as in table 3. Table 4 shows that the precision parameter k for the mean of site means decreases from 8 to 7 after demagnetization, but increases again to 20 after structural restoration; this is clearly again a positive result of Graham's (1949) fold test. Although the scatter of the demagnetized site means ($k = 20$, table 4) is less than the scatter of all site means before demagnetization ($k = 12.8$, table 3), figure 8, *b* shows that the means of these two sets are not significantly different. It follows from this that the remainder of the specimens, with intensities of remanence less than 10^{-4} e.m.u., carry no appreciable secondary magnetization.

The scatter of the site means after demagnetization and structural restoration (fig. 8, *b*) contains almost no contribution from secondary magnetization. Thus it can be accounted for as the sum of contributions from the secular variation of the geomagnetic field at the time of magnetization of the rocks, and from residual uncertainties in structural restoration as discussed above. The data for Tertiary igneous rocks summarized by Cox and Doell (1960, p. 678–685), in which any effects due to deformation have been accounted for in most if not all instances, indicate that a value of k as low as 20 is not unusual. It appears from this that the scatter of site means illustrated

TABLE 4*

RESULTS OF A.C. DEMAGNETIZATION: SITE-MEAN DIRECTIONS OF MAGNETIZATION BEFORE AND AFTER TREATMENT, AND MEAN OF SITE MEANS (*a*) BEFORE TREATMENT, (*b*) AFTER TREATMENT, AND (*c*) AFTER TREATMENT AND STRUCTURAL RESTORATION

Site	*H*	*D*	*I*	*N*	*R*	*k*	α_{95}
2	0	65	81	3	2.9304	29	.....
	100	210	88		2.9883	171	9.5
3	0	75	64	4	3.9565	69	.....
	175	83	73		3.9813	160	7.3
12	0	281	71	5	4.9319	59	.....
	200	228	63		4.9733	150	6.3
16	0	211	41	3	2.9989	1800	.....
	75	209	29		2.9977	870	4.2
17	0	329	88	3	2.9737	76	.....
	125	169	70		2.9380	32	22.1
19	0	208	41	3	2.9715	70	.....
	100	210	30		2.9646	57	16.6
20	0	222	66	7	6.9789	284	.....
	150	210	32		6.8865	53	8.4
Mean of Site Means:							
(*a*)		210	76	7	6.2534	8	.....
(*b*)		204	60		6.1818	7	.....
(*c*)		78	47		6.7033	20	13.8

* H, peak initial demagnetizing field in oersteds; other symbols as in tables 1 and 3.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

in figure 8, *b* may be due in large part to secular variation of the geomagnetic field.

A number of specimens were remeasured after two years' storage in the earth's field in the laboratory; they showed no significant deviations in direction. The partial demagnetization experiments described above have resulted in no significant change in the estimated over-all mean magnetization direction (mean of site means). Furthermore, the magnetization directions of three types of rock (spilitic basalt, albite diabase, and labradorite diabase) are mutually consistent and deviate markedly from the present field direction. The magnetite in these rocks is certainly primary, and the sphene associated with it probably formed during the later stages of the cooling of the rocks. Thus the weight of evidence indicates that the observed directions are largely stable and that they represent magnetization acquired by the rocks prior to their folding and probably of thermoremanent origin.

AGE OF THE FRANCISCAN FORMATION IN WESTERN MARIN COUNTY

Geologic and paleontologic evidence.—The eugeosynclinal assemblage of rocks in western Marin County and on the San Francisco peninsula, including the basic igneous rocks, was originally mapped by Lawson (1914) as the "Franciscan group" and was tentatively considered to be very late Jurassic to early Cretaceous in age. He thought (p. 7) that it was younger than, though only in fault contact with, the granites in the Coast Ranges which, by correlation with the Sierra Nevada granites, were post-Jurassic. Lawson also considered the Franciscan to be unconformably overlain by the Knoxville Formation of early Cretaceous age. Lawson himself pointed out the apparently contradictory nature of this interpretation, and later work has intensified it. Potassium-argon dating (Curtis, Evernden, and Lipson, 1958) has established the late Cretaceous age of the Coast Range granites, and the Knoxville Formation is now considered to be late Jurassic (Tithonian) in age (Irwin, 1957).

The very limited paleontological data relating to the Franciscan Formation have been reviewed by Irwin (1957). He concluded, on the basis of fossils ranging in age from late Jurassic (Tithonian) to early late Cretaceous (Cenomanian), that the Franciscan-Knoxville relation is a facies change obscured by extensive faulting rather than an unconformity (Lawson, 1914) or conformable superposition (Taliaferro, 1943).

Fossils have been described from few localities in Marin County in rocks described as Franciscan. One of these is a *Mantelliceras* of late Cretaceous (Cenomanian) age, from the southern tip of the Marin Peninsula (Hertlein, 1956). Two forms from the vicinity of Novato, approximately 9 miles northeast of the area studied, were reported by Durham and Jones (1959). They are *Buchia piochii* (Gabb) of late Jurassic (Tithonian) age and *Inoceramus schmidti* of late Cretaceous (Campanian) age. A limestone containing late Cretaceous (Cenomanian) foraminifera crops out within 1½ miles of the spilitic bodies (Thalmann, 1943). Although it is lithologically similar to, and contains the same fauna as, the Calera limestone "member" of the "Franciscan group" as mapped by Lawson (1914) on the San Francisco peninsula, this particular limestone exposure is within a sliver tectonically included in the San Andreas fault zone and cannot be used to assign an age to other rocks in Marin County.

An unsuccessful attempt to date the labradorite diabase at site 6 by the potassium-argon method has been reported by Evernden, Savage, Curtis, and James (1964), which gave a minimum age of 65 million years. At the present time it can only be said that, according to the fossil evidence, the age of the units studied could be anything from late Cretaceous (Campanian) to late Jurassic (Tithonian), *i.e.*, from 80 to 140 million years old according to Kulp's (1961) time scale.

PALEOMAGNETIC POLE POSITIONS

Comparison with other data.—To facilitate comparison with other paleomagnetic re-

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

sults, virtual geomagnetic pole positions are given in table 5, as calculated from the mean directions of magnetization after structural restoration (tables 2, 3, and 4). Pole position (*c*) is the best estimate, as it is least influenced by secondary magnetizations; but there are no significant differences among the three. All of the published upper Jurassic and Cretaceous pole positions for North America are also given in table 5. The data are not abundant, but all poles the scatter of directions is large and the paleomagnetic poles are close to the geographic pole. Turonian (upper Cretaceous) igneous rocks in Madagascar give a pole position (Roche *et al.*, quoted by Cox and Doell, 1960) which is not far from that obtained by Larochelle (1962) for the Monteregian Hills (cf. table 5). Finally, the Cretaceous Inkstone red shales of Japan (Nagata *et al.*, quoted by Cox and Doell, 1960) give a pole position about 40° N. and

TABLE 5

VIRTUAL GEOMAGNETIC POLE POSITIONS*

Location	Latitude	Longitude	δ_m	δ_p	Age (10^6 years)
California: Franciscan Formation: (*a*)	29° N.	42° W.	10.3	6.3	(80–140, estimated)
(*b*)	29° N.	44° W.	10.5	6.6	.....
(*c*)	26° N.	50° W.	17.8	11.5	.....
California: Sierra Nevada granites†	70.5° N.	171° E.	9.8	7.5	78–89‡
Quebec: Mt. Megantic§	68.5° N.	171.5° E.	8.9	6.1	113‖
Quebec: Monteregian Hills#	65° N.	157° W.	11.7	9.0	124‖
Arizona: Dakota sandstone**	80° N.	176° W.	21.5	16.0	(Cretaceous)
Utah: Carmel Formation††	80° N.	160° W.	14.0	11.0	(upper Jurassic)

* Results from the Franciscan rocks are calculated (*a*) from the over-all mean direction, (*b*) from the mean of all site-means, and (*c*) from the mean of demagnetized site-means. σ_m and σ_p are the semiaxes of the ovals of 95 per cent confidence, in degrees Other North American results from the Cretaceous and upper Jurassic are included for comparison.

† Currie, Grommé, and Verhoogen (1963).

‡ Curtis *et al.* (1958).

§ Larochelle (1961).

‖ Fairbairn, Faure, Pinson, Hurley, and Powell (1963).

Larochelle (1962).

** Runcorn; quoted by Cox and Doell (1960).

†† Collinson and Runcorn; quoted by Cox and Doell (1960).

are in latitudes higher than 60° and have longitudes between 90° W. and 270° W. The only comparable paleomagnetic data from Europe are from lower Cretaceous sediments in the Wealden area of southern England; in one instance the pole position is not significantly different from the present geographic pole (Nairn, 1960), and in the other the result of the fold test is apparently negative (Wilson, 1959). The paleomagnetism of upper Cretaceous (Huitrinian) and upper Jurassic (Kimmeridgian) sediments from Nequen province in Argentina has been investigated by Creer (1962*b*); in both cases 150° W., in the northeast Pacific. Thus, from all the investigated rocks of upper Jurassic or Cretaceous age from North America, South America, Europe, Africa, and Japan, no paleomagnetic pole position has been obtained which is at all comparable in position to that derived from the Franciscan Formation. The discrepancy amounts to a minimum angular distance of 60°.

From other continents, however, late Mesozoic pole positions have been obtained which are closer to the apparent Franciscan pole. All Mesozoic poles from Australia fall in a rather restricted group in the North

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

Atlantic, at about 50° north latitude (Irving, Robertson, and Stott, 1963). These data are from rocks older than 90 million years, and the same authors have presented cogent arguments for interpreting the systematic difference between these and the corresponding North American and European pole positions as resulting either from a large relative displacement of the Australian continent or from a consistently non-dipolar field throughout the Mesozoic.

The extensively investigated upper Jurassic or Cretaceous Rajmahal traps and Cretaceous to Eocene Deccan traps of India give pole positions in low to intermediate latitudes in the western Atlantic and eastern North America, respectively (cf. Cox and Doell, 1960). These have generally been interpreted as indicating extensive movement of India since the Eocene. Finally, one North American formation, which is demonstrably lower to lower-middle Eocene, gives a pole position with the same longitude and 10° higher latitude than the Franciscan pole; this is the Siletz River volcanics of western Oregon (Cox, quoted by Cox and Doell, 1960). On this basis, it might be supposed that the geomagnetic pole migrated southward into the Atlantic Ocean and back on one or more occasions during the late Mesozoic and Eocene. If such movements occurred, they probably were several in number, because the Siletz River volcanics and the Franciscan Formation in Marin County differ in age by at least 30 million years and probably much more, and the Rajmahal and Deccan traps apparently represent an even greater span of time which includes at least some of the North American results mentioned above. To explain these anomalous pole positions in terms of a single excursion of the geomagnetic pole would require that the Franciscan rocks and the Rajmahal and Deccan traps all be younger than about 80 million years, and that the departure persisted from that time until well into the Eocene. There are no other paleomagnetic data which support such an event during that period, nor are there anv analogous paleomagnetic data from any other part of geologic time.

CONCLUSIONS

Paleomagnetic pole position.—The geomagnetic pole position implied by the paleomagnetic data from the Franciscan Formation in western Marin County lies in the middle of the Atlantic Ocean at about 30° north latitude. This is grossly at variance with all other paleomagnetic data reported for upper Jurassic and Cretaceous rocks in the northern hemisphere. Three hypotheses may be advanced to account for the discrepancy:

1. Tectonic rotation of all three igneous bodies as a single unit, 70° clockwise around a vertical axis occurred. If such a movement occurred, it must have been prior to the folding of the rocks, and therefore in all likelihood prior to the formation of the San Andreas fault. Thus the correspondence of this rotation to the sense of a hypothetical shear couple responsible for the observed displacement on the San Andreas fault appears to be fortuitous. One might speculate that the rotation occurred while the Franciscan geosyncline was being filled and was in a highly mobile state. We wish to emphasize, however, that there is no evidence other than the magnetic data in support of any such deformation.

2. The non-dipole part of the geomagnetic field was temporarily predominant. Considering the nature of the present field, this appears unlikely in the case of the Franciscan rocks. The non-dipole part of the present field is responsible for virtually all of the secular variation. The Franciscan rocks sampled cover an appreciable stratigraphic thickness, probably equivalent to at least several thousand years; thus if the secular variation field had been dominant at that time the observed consistency of magnetization directions from site to site would not have been found. There is no other paleomagnetic evidence from single continents which might indicate that the earth's field possessed a stable, non-varying,

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

non-dipole configuration for any appreciable length of geologic time.

3. There were one or more excursions of the geomagnetic pole into the Atlantic Ocean, assuming an essentially dipolar field. A limited amount of paleomagnetic data from rocks in India and Oregon suggests that this may have happened, but it was shown above that because of the span of geologic time which appears to be involved, more than one such excursion is implied. Evaluation of this last possibility must await (*a*) a more accurate determination of the age of these Franciscan rocks, (*b*) other paleomagnetic data from upper Cretaceous and lower Eocene rocks in North America, and possibly (*c*) radiometric age determinations of the Rajmahal and Deccan traps of India.

Structure of spilitic bodies.—The magnetic data from the samples collected on the east side of the northern spilitic body are not significantly divergent from those collected from the west side of the same body, nor are the directions of magnetization of the coarse-grained rocks distinguishable from those of the fine-grained rocks. Therefore the magnetic measurements do not substantiate the interpretation of a synclinal structure in the northern volcanic body; instead, the uniformity of directions of magnetization of the rocks is what would be expected within a homoclinal sequence and is in agreement with the uniformity of the pillow shapes. The average directions of magnetization of samples from the central and southern bodies agree with those from the northernmost body; thus the field interpretation that the two southern volcanic bodies are also homoclinal and dip to the northeast is reinforced.

We conclude from this that the 15-mile-long belt of basic igneous rocks, composed of three separate bodies, is a relatively uniform upright homoclinal sequence dipping northeastward. This volcanic zone is the most continuous mappable stratigraphic unit in the Franciscan Formation of western Marin County. We suggest that the remanent magnetization of similar volcanic rocks elsewhere within the Franciscan Formation may profitably be investigated with the objectives of gaining both paleomagnetic and structural information.

ACKNOWLEDGMENTS.—We wish to thank Professors John Verhoogen and G. H. Curtis for advice and encouragement and Dr. Allan Cox of the U.S. Geological Survey for critical reading of an earlier draft of the manuscript. This work was supported in part by grants from the National Science Foundation and the Air Force Cambridge Research Laboratories, which are gratefully acknowledged.

REFERENCES CITED

CARLISLE, DONALD, 1963, Pillow breccias and their aquagene tuffs, Quadra Island, British Columbia: Jour. Geology, v. 71, p. 48–71.

COX, ALLAN, 1961, Anomalous remanent magnetization of basalt: U.S. Geol. Survey Bull. 1083-E, p. 131–160.

——— and DOELL, R. R., 1960, Review of paleomagnetism: Geol. Soc. America Bull., v. 71, p. 645–768.

CREER, K. M., 1959, A. C. demagnetization of unstable Triassic Keuper marls from S.W. England: Geophys. Jour., v. 2, p. 261–275.

——— 1962*a*, Paleomagnetism of the Serra Geral formation: *ibid.*, v. 7, p. 1–22.

——— 1962*b*, Paleomagnetic data from South America: Jour. Geomagnetism and Geoelectricity, v. 13, p. 154–165.

CURRIE, R. G., GROMMÉ, C. S., and VERHOOGEN, J., 1963, Remanent magnetization of some upper Cretaceous granitic plutons in the Sierra Nevada, California: Jour. Geophys. Research, v. 68, p. 2263–2279.

CURTIS, G. H., EVERNDEN, J. F., and LIPSON, J., 1958, Age determination of some granitic rocks in California by the potassium-argon method: Calif. Division of Mines and Geology Spec. Rept. 54, 16 p.

DESBOROUGH, G. A., 1963, Mobilization of iron by alteration of magnetite-ulvöspinel in basic rocks in Missouri: Econ. Geology, v. 58, p. 332–346.

DURHAM, J. W., and JONES, D. L., 1959, Fossil occurrences bearing on the Franciscan problem in central California (Abs.): Geol. Soc. America Bull., v. 70, p. 1716.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

EVERNDEN, J. F., SAVAGE, D. E., CURTIS, G. H., and JAMES, G. T., 1964, Potassium-argon dates and the Cenozoic mammalian chronology of North America: Am. Jour. Sci., v. 262, p. 145–198.

FAIRBAIRN, H. W., FAURE, G., PINSON, W. H., HURLEY, P. M., and POWELL, J. L., 1963, Initial ratio of strontium 87 to strontium 86, whole-rock age, and discordant biotite in the Monteregian Igneous Province, Quebec: Jour. Geophys. Research, v. 68, p. 6515–6522.

FISHER, R. A., 1953, Dispersion on a sphere: Royal Soc. London, Proc. ser. A, v. 217, p. 295–305.

GLUSKOTER, H. J., 1964, Orthoclase distribution and authigenesis in the Franciscan Formation in a portion of western Marin County, California: Jour. Sed. Petrol., v. 34, p. 335–343.

GRAHAM, J. W., 1949, The stability and significance of magnetism in sedimentary rocks: Jour. Geophys. Research, v. 54, p. 131–167.

GRAHAM, K. W. T., 1961, The re-magnetization of a surface outcrop by lightning currents: Geophys. Jour. v. 6, p. 85–102.

HERTLEIN, L. G., 1956, Cretaceous ammonite of Franciscan group, Marin County, California: Am. Assoc. Petroleum Geologists Bull., v. 40, p. 1985–1988.

———, ROBERTSON, W. A., and STOTT, P. M., 1963, The significance of the paleomagnetic results from Mesozoic rocks of eastern Australia: Jour. Geophys. Research, v. 68, p. 2313–2317.

IRVING, E., STOTT, P. M., and WARD, M. A., 1961, Demagnetization of igneous rocks by alternating magnetic fields: Philos. Mag., v. 6, p. 225–241.

IRWIN, W. P., 1957, Franciscan group in Coast Ranges and its equivalents in Sacramento Valley, California: Am. Assoc. Petroleum Geologists Bull., v. 41, p. 2284–2297.

JENNINGS, C. W., and BURNETT, J. L., 1961, San Francisco Sheet: Geologic map of California, Olaf P. Jenkins edition, Calif. State Div. Mines and Geology.

KULP, J. L., 1961, Geologic time scale: Science, v. 133, p. 1105–1114.

LAROCHELLE, A., 1961, Application of paleomagnetism to geological correlation: Nature, v. 192, p. 37–39.

——— 1962, Paleomagnetism of the Monteregian Hills, southeastern Quebec: Geol. Survey Canada Bull. 79, 43 p.

LAWSON, A. C., 1914, San Francisco Folio: U.S. Geol. Survey Geol. Atlas, Folio 193.

NAGATA, TAKESI, 1961, Rock magnetism: rev. ed., Tokyo, Maruzen, 350 p.

NAIRN, A. E. M., 1960, Paleomagnetic results from Europe: Jour. Geology, v. 68, p. 285–306.

SHROCK, R. R., 1948, Sequence in layered rocks: New York, McGraw-Hill Book Co., 507 p.

SMITH, J. V., 1956, The powder patterns and lattice parameters of plagioclase feldspars, I. The soda rich plagioclases: Mineralog. Mag., v. 31, p. 47–68.

TALIAFERRO, N. L., 1943, Franciscan-Knoxville problem: Am. Assoc. Petroleum Geologists Bull., v. 27, p. 109–228.

THALMANN, H. E., 1943, Upper Cretaceous age of the "Franciscan" limestone near Laytonville, Mendocino County, California (Abs.): Geol. Soc. America Bull., v. 54, p. 1827.

WATSON, G. S., and IRVING, E., 1957, Statistical methods in rock magnetism: Royal Astron. Soc., Monthly Notices, Geophys. Supp., v. 7, p. 289–300.

WEAVER, C. E., 1949, Geology of the Coast Ranges immediately north of the San Francisco Bay region, California: Geol. Soc. America Memoir 35, 242 p.

WILSON, R. L., 1959, Remanent magnetism of late Secondary and early Tertiary British rocks: Philos. Mag., v. 4, p. 750–755.

This content downloaded from 205.208.116.024 on September 21, 2017 20:25:39 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).